

Homework 2 — Nash Equilibrium Algorithms

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Abstract

This report presents two algorithms from game theory rendered as computer programs. Problem 1 finds all pure Nash Equilibria (NE) of an n -player normal-form game via exhaustive best-response checking. Problem 2 finds the completely mixed NE of a 2×2 game in closed form, with a discussion of how the approach generalizes to more strategies. Both programs take all inputs from the keyboard, supporting either manual payoff entry or randomized generation, and both are validated against well-known games.

1 Problem 1: Pure Nash Equilibria of an n -Player Game

1.1 Algorithm

The program `prob1.py` reads the number of players $n \leq 10$, the number of strategies of each player, and the payoffs of all players at every strategy profile (manually or by randomization within a chosen range). A profile $s = (s_1, \dots, s_n)$ is a pure NE iff no player can strictly increase his payoff by unilaterally switching strategy. The algorithm enumerates every one of the $S = \prod_i s_i$ profiles and, for each, checks every player's alternative strategies:

```
for profile in all_profiles:
    is_ne = True
    for player in players:
        for alt in player.strategies (alt != profile[player]):
            if payoff(player, profile with alt) > payoff(player, profile):
                is_ne = False
    if is_ne: report profile as a pure NE
```

This is exhaustive and exact: it correctly answers existence (equilibrium list empty $\iff$ no pure NE) and finds every NE. Cost is $O(n \cdot S \cdot \max_i s_i)$ payoff comparisons – essentially linear in the size of the payoff table itself.

1.2 Simulation results

Scenario	Size	Payoffs	Pure NE found
Prisoner's Dilemma	2 players, 2×2	Manual	(D,D) $\rightarrow$ (1,1) — 1 NE
Battle of the Sexes	2 players, 2×2	Manual	(Mall,Mall) $\rightarrow$ (2,1), (Football,Football) $\rightarrow$ (1,2) — 2 NE
Matching Pennies	2 players, 2×2	Manual	None — 0 NE
Random game	3 players, 2^3	Random [0, 5]	$(s_2, s_2, s_2) \rightarrow (5, 5, 5)$ — 1 NE
Random game	2 players, 3×3	Random [0, 10]	$(s_3, s_3) \rightarrow (10, 8)$ — 1 NE

2 Problem 2: Completely Mixed NE of a 2×2 Game

2.1 Derivation

Let player 1 mix $(p, 1 - p)$ and player 2 mix $(q, 1 - q)$ over (S_1, S_2) , with payoff matrices A (player 1) and B (player 2). A completely mixed NE requires each player to be exactly indifferent between his two strategies. Equating player 2's expected payoffs pins down p ; equating player 1's pins down q :

$$p = \frac{b_{22} - b_{21}}{b_{11} - b_{12} - b_{21} + b_{22}}, \quad q = \frac{a_{22} - a_{12}}{a_{11} - a_{12} - a_{21} + a_{22}}$$

A completely mixed NE exists iff both $p, q \in (0, 1)$; a zero denominator means the indifference condition is degenerate and `prob2.py` reports no completely mixed NE.

2.2 Simulation results

Scenario	Payoffs	Result
Battle of the Sexes	Manual	$p = 0.667, q = 0.333$ — mixed NE
Prisoner's Dilemma	Manual	No mixed NE (matches theory)
Matching Pennies	Manual	$p = q = 0.500$ — mixed NE
Random game	Random $[0, 10]$	No mixed NE (p, q outside $(0, 1)$)

2.3 Generalization to more strategies

For an $m_1 \times \dots \times m_n$ game, a completely mixed NE requires every player indifferent across all m_i of his strategies given the others' mixes. Since expected payoff is multilinear in the mixing probabilities, this gives $m_i - 1$ linear equations per player, plus one normalization ($\sum$ probabilities = 1); stacking these over all players yields a linear system solvable with standard linear algebra (e.g. `numpy.linalg.solve`). A solution is a genuine completely mixed NE only if every probability lies strictly in $(0, 1)$; with more than two strategies one should also confirm no pure deviation improves any payoff. When the fully-mixed system has no valid solution, general methods such as Lemke–Howson (2 players) or support enumeration locate other equilibria.